Product Requirements Document

Intelligent AI-Powered PC Onboarding Assistant (Abhi)

Document Control

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1. Executive Summary

1.1 Product Vision

The Intelligent AI-Powered PC Onboarding Assistant (codename: **Abhi**) revolutionizes the first-time user experience for AI PCs by providing a conversational, automated setup process that eliminates friction for non-technical users. Abhi transforms PC onboarding from a technical burden into a seamless, personalized experience that demonstrates the value of AI from the first moment of device ownership.

1.2 Customer Pain Points

Current State Problems:

1. **Complexity Overwhelms Non-Technical Users**
    - Traditional PC setup presents 50+ configuration decisions
    - Technical jargon (BIOS, drivers, firmware) confuses mainstream consumers
    - Users uncertain which settings to choose, leading to suboptimal configurations
    - Fear of "doing it wrong" creates anxiety and hesitation
2. **Time-Consuming Manual Setup**
    - Average OOBE (Out-of-Box Experience) takes 45-60 minutes
    - Additional 2-3 hours for app installation, file migration, peripheral setup
    - Multiple restarts interrupt workflow and test user patience
    - Users abandon setup mid-process, leaving PCs partially configured
3. **Data Migration Friction**
    - Users struggle transferring files from old PC to new PC
    - Cloud sync setup confusing for non-technical users
    - Network transfer requires technical knowledge (IP addresses, file shares)
    - USB transfer prone to user error (wrong folders, incomplete copies)
4. **Peripheral Configuration Challenges**
    - Multi-monitor setup requires manual display arrangement
    - Driver updates buried in manufacturer websites or forgotten entirely

    - Bluetooth pairing frustrates users with cryptic error messages
    - External devices (printers, webcams) require separate setup workflows
5. **Generic One-Size-Fits-All Experience**
    - Traditional OOBE doesn't adapt to user persona (gamer vs. business user vs. student)
    - Recommends irrelevant apps, wasting time and storage
    - No context-aware guidance based on user's actual needs
    - Users forced to manually hunt for apps and settings post-setup
6. **High Support Burden**
    - Significant portion of support calls in first 30 days are setup-related
    - Common issues: driver problems, app installation failures, data migration questions
    - Support teams spend time on routine, automatable tasks
    - Negative first impression impacts brand perception and satisfaction

1.3 Value Proposition

What Abhi Delivers:

For Consumers:

- **Effortless Setup in 30 Minutes**: Conversational AI guides entire process—no technical knowledge required. Apps and configuration completed behind the scenes in the background for a seamless experience
- **Personalized Experience**: AI detects user persona and recommends relevant apps, settings automatically
- **Hands-Free Migration**: Automated file transfer from old PC via cloud, network, or USB with intelligent conflict resolution
- **Plug-and-Play Peripherals**: Automatic driver updates, multi-monitor configuration, and Bluetooth pairing
- **Peace of Mind**: Session persistence means interruptions (crashes, power loss) don't reset progress

For IT Managers & Enterprises:

- **Support Cost Reduction**: Automated guidance eliminates routine setup support tickets
- **Standardized Deployments**: Consistent setup process across fleet ensures compliance and reduces configuration drift
- **Faster Time-to-Productivity**: Employees operational in 30 minutes vs. 3+ hours
- **Deployment Insights**: Telemetry reveals common setup issues, enabling proactive improvements
- **Reduced IT Overhead**: Less hands-on involvement needed for device provisioning

For OEMs & PC Manufacturers:

- **Customer Satisfaction Lift**: Reduced setup friction increases Net Promoter Score (NPS)
- **Reduced Returns**: Proper initial configuration decreases "this doesn't work" returns
- **Brand Loyalty**: Positive first impression strengthens customer relationship
- **Category Leadership**: Showcase AI capabilities at moment of highest engagement

1.4 Business Objectives

- **Reduce customer support calls by 20%** through automated setup guidance and assistance
- **Increase first-week device engagement by 40%** through setup completed via assistant
- **Achieve 90%+ setup completion rate** through guided, interruption-resistant experience
- **Differentiate AI PC value proposition** by showcasing AI capabilities immediately post-purchase

1.5 Success Metrics (KPIs)

Metric	Target	Measurement Method
Setup Completion Rate	≥ 90%	Telemetry: Session start vs. completion events
Average Setup Time	8-12 minutes	Session duration telemetry
Customer Satisfaction (CSAT)	≥ 4.5/5.0	Post-setup survey

Metric	Target	Measurement Method
Support Ticket Reduction	20% decrease	Support ticket volume analysis (30-day comparison)
App Installation Success	≥ 95%	Installation completion tracking

1.6 Investment & ROI

To be determined.

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2. Market Context & Target Audience

2.1 Market Opportunity

To be determined.

2.2 Target Customer Segments

Primary: Consumer AI PC Buyers

- Demographics: Ages 25-65, moderate-to-high income, college-educated
- Tech proficiency: Low to moderate
- Pain points: Setup complexity, data migration anxiety, feature uncertainty

Secondary: SMB & Enterprise Deployments

- IT administrators deploying AI PCs to knowledge workers
- Need: Standardized, repeatable setup with centralized management

2.3 Primary Persona: Sarah - The Mainstream Consumer

Attribute	Details
Demographics	Age 35, marketing manager, suburban parent of two, moderate tech literacy
Tech Environment	Upgrading from 5-year-old laptop, uses cloud services (OneDrive, Gmail), owns smartphone/tablet
Goals	Get new PC running quickly; transfer photos and documents; understand what makes AI PC special
Pain Points	- Overwhelmed by technical jargon and setup wizards - Doesn't know which apps to install - Worried about losing files during migration - Uncertain how to use AI features - Fear of missing critical setup steps
Success Criteria	PC fully configured with files transferred and essential apps installed

Attribute	Details
	within 30 minutes, without calling support
Quote	"I just want my PC to work like my old one, but better—without spending hours figuring it out."

2.4 Competitive Landscape

Solution	Approach	Limitations	Our Differentiation
Traditional Windows OOBE	Static wizard, predetermined steps	Not personalized, requires technical knowledge, no automation	AI-driven conversation adapts to user needs
Apple Migration Assistant	Automated data transfer between Apple devices	Limited to Apple ecosystem, no personalization, no AI capabilities	Cross-platform support with intelligent automation
PC Transfer Tools (PCmover, Laplink)	File transfer utilities	Data transfer only, no system configuration, paid software	End-to-end setup automation beyond data migration
Dell/HP Setup Tools	Vendor-specific utilities	Limited to hardware config, not persona-aware, fragmented UX	Unified, AI-powered experience across all OEMs

2.5 Value Proposition

Core Promise: "Your AI PC sets itself up"

- **For Users**: Transform from "I don't know where to start" to "My PC is ready to use" in under 15 minutes
- **For OEMs**: Differentiate AI PCs with tangible value; reduce support costs by 20%; increase customer satisfaction and NPS scores
- **For the Business**: Establish AI PC category leadership; create platform for ongoing AI feature adoption; generate valuable user insights

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3. Product Overview

3.1 Product Description

Abhi (AI-Based Helper Interface) is an intelligent onboarding assistant that launches immediately after Windows OOBE completion. Using conversational AI powered by on-

device small language models (SLMs), Abhi conducts a personalized interview to understand user needs, then automatically configures the system, installs applications, updates firmware, and migrates data with minimal user intervention.

Key Differentiators:

- First AI-native PC setup experience
- Multi-modal interaction (text, voice, image)
- Persona-aware recommendations
- Background automation during conversation
- Privacy-first, on-device AI processing

3.2 Core Capabilities

Capability	Description	Business Value
Multi-Modal Interaction	Chat, voice, and image-based communication	Accessibility, user preference accommodation, inclusivity
Persona Detection	AI identifies user type (gamer, student, professional, creator, casual)	Tailored recommendations drive satisfaction and reduce cognitive load
Automated Configuration	System settings, display, peripherals, power management	Time savings, error reduction, professional setup quality
Software Provisioning	Intelligent app recommendations and automated installation	Feature discovery, productivity readiness, reduced abandonment
Data Migration	Cloud sync, network transfer, or USB-based migration	Continuity preservation, anxiety reduction, faster productivity
Update Management	Background firmware, driver, and OS updates	Security posture, stability, optimal performance from day one
Session Persistence	Automatic save/resume functionality	Interruption resilience, user confidence, completion rate improvement

3.3 User Journey Flow

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│                      Windows OOBE Completes                  │
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┌──────────────────────────────────────────────────────────────┐
│                 First Login (Desktop Shell Loads)            │
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                    ▼ (< 10 seconds)
┌──────────────────────────────────────────────────────────────┐
│                  Abhi Auto-Launches (Fullscreen)             │
│                                                              │
│  ┌──────────────────────────────────────────────────────────┐│
│  "Hi! I'm Abhi, your AI assistant. Let me help you          │
│   set up your new PC."                                       │
│                                              [Skip Setup →]  │
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                ▼                   ▼
        ┌──────────────────┐  ┌──────────────────┐
        │ Power User       │  │ New User         │
        │ Clicks Skip      │  │ Continues        │
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                │                   │
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        │ Confirmation     │  │ Personalized Interview Begins    │
        │ Dialog →         │  │ (5-15 minutes)                   │
        │ Exit & Flag      │  │                                  │
        └──────────────────┘  │ • Usage patterns questions       │
                              │ • Persona detection              │
                              │ • Multi-modal interaction        │
                              │ • Background automation starts   │
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                              │ Data Migration Phase             │
                              │ (if applicable)                  │
                              │                                  │
                              │ • Detect previous device         │
                              │ • Offer migration options        │
                              │ • Execute transfer               │
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                              │ Configuration Review & Approval  │
                              │                                  │
                              │ • Show recommended apps          │
                              │ • Display detected settings      │
                              │ • Request user confirmation      │
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```
Final Setup Execution

• Install approved apps
• Apply configurations
• Complete pending updates
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Setup Complete!

• Success message
• Optional restart if needed
• Quick tips for AI features
• Satisfaction survey
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4. Technical Requirements

4.1 System Architecture

4.1.1 High-Level Architecture Diagram

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User Interface Layer

Technology: Per company branding and tech guidelines
• Chat Interface  • Voice I/O  • Image Upload
• Progress Visualization  • Consent Dialogs

IPC / Named Pipes / A2A / MCP

Application Logic Layer

AI Engine (On-Device SLM)
• Natural Language Understanding
• Persona Classification
• Intent Recognition
• NPU/GPU Inference (Fallback: CPU)

Orchestration Service (Windows Service)
• Task Queue Manager
• State Machine Controller
• Session Persistence
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    - Context window: 4K tokens (sufficient for multi-turn conversation)
- **Fallback**: Cloud-based AI service for complex queries (requires user consent, logged telemetry)

2. User Interface Layer

- **Technology**: Follow company branding and technology guidelines
- **Accessibility**: WCAG 2.1 AA compliance, full Narrator/JAWS support, keyboard navigation
- **Features**:
    - Real-time chat interface with markdown rendering
    - Push-to-talk voice input with visual waveform
    - Drag-drop image upload with preview
    - Progress indicators for background tasks
    - Consent dialogs with clear explanations

3. Orchestration Service

- **Implementation**: Windows Service (runs as SYSTEM with privilege management)
- **Responsibilities**:
    - Task queue management (priority-based scheduling)
    - Background automation coordination
    - System resource monitoring and throttling
    - Restart coordination (schedule post-setup if needed)
- **Communication**: Named pipes for IPC with UI process, A2A, and MCP
- **Logging**: ETW (Event Tracing for Windows) or other alternative technology for diagnostic telemetry

4. Specialized Managers

Update Manager

- Windows Update API integration
- OEM firmware update APIs (Dell Command | Update)

- Driver update via Windows Update and OEM catalogs
- Smart restart scheduling (avoid interruption during setup)

App Install Manager

- Microsoft Store COM API for Store apps
- Winget CLI integration for open-source/third-party apps
- Silent installer orchestration (MSI, MSIX, EXE)
- Installation success/failure handling with retry logic
- App catalog: 200+ vetted applications across 12 categories

Device Manager

- WMI queries for hardware enumeration
- PnP (Plug and Play) subsystem integration
- Multi-monitor detection and optimal configuration
- Peripheral pairing assistance (Bluetooth, USB)
- Driver currency validation

Migration Engine

- OneDrive File On-Demand configuration
- Google Drive, Dropbox client integration
- SMB network share discovery (for local transfer)
- USB device file enumeration and selective transfer
- Conflict resolution (duplicate file handling)

4.2 Performance Requirements

Metric	Target	Rationale
Application Launch Time	< 10 seconds	First impression critical; user should not perceive delay
AI Response Latency (On-Device)	< 2 seconds	Conversational flow requires near-real-time response
AI Response Latency (Cloud Fallback)	< 5 seconds	Acceptable for complex queries requiring cloud processing

Metric	Target	Rationale
Memory Footprint (Active Session)	< 500 MB RAM (excluding AI model)	Minimize resource consumption during setup
AI Model Memory Footprint	Up to 2 GB RAM during inference	Additional memory required for AI model execution
Storage Footprint (Installation)	< 2 GB (excluding AI models)	Application binaries and cached assets
AI Model Storage Footprint	Additional storage TBD based on model selection	AI models can require significant disk space
CPU Usage (Idle)	< 5%	Background service should be imperceptible
CPU Usage (Active Inference)	< 20% average	Ensure system remains responsive during AI processing
Network Bandwidth (Cloud Sync)	< 10 Mbps sustained	Accommodate typical residential internet speeds
Battery Impact (Laptop)	< 5% drain	Setup should not significantly deplete battery

4.3 Scalability & Reliability

- **Concurrent Sessions**: Support 1M+ simultaneous onboarding sessions globally (cloud analytics endpoint)
- **Availability**: 99.9% uptime for optional cloud services (AI fallback, telemetry ingestion)
- **Graceful Degradation**: Core functionality operates offline; cloud-dependent features degrade gracefully with clear user messaging
- **Error Recovery**: Automatic retry logic for transient failures; comprehensive error logging for diagnostics
- **Crash Recovery**: Session state persisted every 30 seconds; automatic resume on process restart

4.4 Security & Privacy Requirements

Data Protection

- **Encryption at Rest**: AES-256 encryption for local encrypted database
- **Encryption in Transit**: TLS 1.3 for all network communications
- **Credential Handling**: Windows Credential Manager for cloud account tokens; no plaintext password storage

- **PII Handling**: Minimize PII collection to only what is necessary for device setup and runtime

Privacy Principles

- **Data Minimization**: Collect only setup-essential metadata (device ID, session ID, persona classification, app selections)
- **Transparency**: Clear disclosure in EULA and first-run consent dialog
- **User Control**: Users can manually clear all Abhi data via Settings or PowerShell cmdlet
- **No Cross-User Data Sharing**: Each user's data isolated; no aggregation for training without consent
- **Right to Deletion**: GDPR/CCPA data subject rights supported via OEM support channels

Security Hardening

- **Code Signing**: All binaries signed with EV code signing certificate
- **Integrity Validation**: Hash verification for downloaded apps and updates
- **Least Privilege**: UI process runs as user; service runs as SYSTEM with minimal permissions
- **Sandboxing**: Application processes run in appropriate security contexts
- **Supply Chain Security**: App recommendations limited to Microsoft Store, official vendor sites, and vetted winget sources

4.5 Platform Requirements

Operating System: Windows 11 25H2 or later

Hardware Minimum Requirements:

- **CPU**: 8-core Intel/AMD processor (2019 or newer)
- **RAM**: 8 GB minimum (16 GB recommended)
- **Storage**: 5 GB free space (additional space required for AI models)
- **NPU**: Optional (40+ TOPS for optimal AI performance; GPU/CPU fallback supported)
- **GPU**: Discrete GPUs supported - Nvidia RTX 2000 (Blackwell) or higher
- **Network**: Optional internet connection (offline mode available with degraded features)

Hardware Recommended Specifications:

- **CPU**: 12+ cores with NPU integration (Intel Core Ultra, AMD Ryzen AI)
- **RAM**: 16 GB or more
- **Storage**: NVMe SSD with 10+ GB free
- **NPU**: 40+ TOPS (Intel AI Boost)
- **GPU**: Nvidia RTX 2000 (Blackwell) or higher for enhanced AI performance
- **Network**: Broadband connection (10+ Mbps)

4.6 Integration Points & APIs

External System	Purpose	API/Interface	Criticality
Windows Update	OS patches, cumulative updates	WUA API, PowerShell cmdlets	Critical
Microsoft Store	Consumer app installation	ms-windows-store: protocol, Store COM API	High
Winget	Package manager for non-Store apps	Winget CLI, REST APIs	High
OEM Update Services	Firmware, BIOS, driver updates	Dell Command API	Medium
OneDrive	Cloud file sync and migration	OneDrive Sync Client, Microsoft Graph API	Medium
Google Drive	Cloud file sync (optional)	Google Drive API, OAuth 2.0	Low
Dropbox	Cloud file sync (optional)	Dropbox API, OAuth 2.0	Low
Speech Recognition	Voice input processing	Text-to-speech and speech recognition APIs	Medium
Text-to-Speech	Voice output	Text-to-speech synthesis	Low
Vision Services	Image analysis (desk setup)	Vision and image recognition APIs	Low

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5. Detailed Functional Requirements

5.1 Epic 1 - Initial Launch and Opt-out

Business Value: Ensure seamless activation while respecting power user preferences
Priority: P0 (Must-Have)

User Story 1.1.1 - Auto-launch After Login

Priority: P0 (Must-Have)

As a new PC owner completing Windows OOBE
I want Abhi to launch automatically after my first login
So that I can immediately begin setting up my PC without searching for setup tools

Acceptance Criteria:

- **AC 1.1.1.1**: Abhi application launches within 10 seconds of first desktop load (measured from Shell Ready event to UI visible)
- **AC 1.1.1.2**: Launch is triggered only once per device (registry flag: `HKLM\SOFTWARE\Abhi\FirstRunComplete = 1`)
- **AC 1.1.1.3**: UI appears in foreground and focused on primary display
- **AC 1.1.1.4**: If launch fails, error logged to Event Viewer (Application log, source: `Abhi`, Event ID 1001) with actionable message
- **AC 1.1.1.5**: Launch does not interfere with other startup programs (non-blocking, priority: below-normal)

Technical Implementation Notes:

- **Trigger Mechanism**: Task Scheduler task with trigger type `AtLogon`, delay 15 seconds post-logon
- **Launch Script**: PowerShell wrapper checks registry flag before launching executable
- **Error Handling**: Try-catch with fallback to manual launch shortcut in Start Menu
- **Dependencies**: .NET 8 Runtime or other runtime as determined by technology selection
- **Telemetry**: Log launch event with timestamp, success/failure, launch duration

Testing Scenarios:

1. Fresh OOBE on new device → Abhi launches
2. `Subsequent logins → Abhi does not launch
3. Simulated launch failure → Error logged, user notified
4. Multi-user device → Abhi launches once per user
5. Performance: Launch time < 10s on minimum-spec hardware

User Story 1.1.2 - Power User Opt-out

Priority: P0 (Must-Have)

As a power user or IT professional
I want the ability to skip the guided setup assistant
So that I can configure my system manually using familiar tools

Acceptance Criteria:

- **AC 1.1.2.1**: "Skip Setup" button visible and prominent on initial screen (minimum 120x48px touch target)
- **AC 1.1.2.2**: Clicking skip displays confirmation dialog:
  - Title: "Skip Automated Setup?"
  - Message: "You'll need to manually configure system settings, install apps, and migrate data. You can always access Abhi later from the Start Menu."
  - Buttons: [Continue Setup] [Skip]
- **AC 1.1.2.3**: Confirming skip closes Abhi and sets registry flag `HKLM\SOFTWARE\Abhi\UserOptedOut = 1`
- **AC 1.1.2.4**: Keyboard shortcut (Alt+Esc) provides alternate skip method for accessibility
- **AC 1.1.2.5**: Telemetry logged: skip event with device type, user profile metadata (anonymized)
- **AC 1.1.2.6**: After skip, Start Menu shortcut remains available ("Set up my PC with Abhi")

Technical Implementation Notes:

- **UI Design**: Skip button styled as secondary action (gray, not blue primary color)
- **Confirmation Dialog**: Use Windows Task Dialog API (`TaskDialogIndirect`) for native look/feel
- **Registry Management**: Use `Set-ItemProperty` cmdlet for registry writes with error handling
- **Telemetry**: Anonymized event with device SKU, OEM ID, skip reason (if provided)

Testing Scenarios:

1. Click skip → Confirmation dialog appears
2. Confirm skip → Abhi closes, registry flag set
3. Alt+Esc keyboard shortcut → Skip confirmation dialog
4. Cancel skip dialog → Return to Abhi main screen
5. Post-skip: Manual launch from Start Menu works

5.2 Epic 2 - Personalized Interview & Session Flow

Business Value: Adaptive, contextual setup maximizes user satisfaction and reduces abandonment
Priority: P0 (Must-Have)

User Story 1.2.1 - Multi-Modal Interaction

Priority: P1 (High Priority)

As a user with diverse accessibility needs and communication preferences
I want to interact with Abhi using text chat, voice commands, or image uploads
So that I can communicate in the way that's most comfortable for me

Acceptance Criteria:

- **AC 1.2.1.1**: Text chat interface with:
  - Rich text support (markdown rendering)
  - Typing indicators while AI processes
  - Message history scrollback (up to 50 messages)
  - Copy message button for each AI response
- **AC 1.2.1.2**: Voice input:
  - Push-to-talk button (hold to record, release to send)
  - Visual waveform during recording
  - Automatic silence detection (2-second trailing silence → send)
  - Transcription displayed in chat before sending (editable)
- **AC 1.2.1.3**: Voice-to-text accuracy ≥ 95% (measured on standard test corpus in English, Spanish, French, German, Chinese)

- **AC 1.2.1.4**: Text-to-speech output:
    - Toggle button to enable/disable voice responses
    - Configurable voice (male/female/neutral) and speed (0.75x - 1.5x)
    - Natural prosody (not robotic monotone)
- **AC 1.2.1.5**: Image upload:
    - Drag-drop or file picker
    - Supported formats: JPG, PNG, HEIC, WebP (max 10MB per image)
    - Preview thumbnail in chat after upload
    - AI analysis for desk setup photos (detect monitors, peripherals)
- **AC 1.2.1.6**: Image analysis accuracy ≥ 80% for common peripherals (monitor, keyboard, mouse, webcam, speakers)

Technical Implementation Notes:

- **Speech Recognition**: Speech recognition API with continuous recognition mode and custom grammar for setup-related terms
- **TTS Engine**: Text-to-speech synthesis with SSML support for prosody control and voice selection
- **Image Analysis**: Vision API for on-device object detection; fallback to cloud-based Computer Vision API (with user consent) for complex scenes
- **Accessibility**: Full ARIA labels, keyboard navigation (Tab, Space, Enter), high contrast theme support, Narrator announcements for state changes
- **Performance**: Voice input latency < 500ms, TTS output latency < 1s, image analysis < 3s

Testing Scenarios:

1. Text chat: Send message, receive response with < 2s latency
2. Voice input: Record audio, verify transcription accuracy
3. Voice output: Enable TTS, verify speech plays with correct prosody
4. Image upload: Drag-drop desk photo, verify peripheral detection
5. Accessibility: Navigate entire UI with keyboard only, verify screen reader compatibility

6. Error handling: Network failure during cloud API call → graceful degradation with message

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User Story 1.2.2 - Persona Detection

Priority: P0 (Must-Have)

As a user describing how I plan to use my PC
I want Abhi to understand my usage profile and recommend appropriate apps and settings
So that my PC is configured for my specific needs without trial-and-error

Acceptance Criteria:

- **AC 1.2.2.1**: AI classifies user into one of 5 primary personas within first 3-5 conversation turns:
  - Gamer
  - Student/Educator
  - Professional/Business
  - Content Creator
  - Casual/Home User
- **AC 1.2.2.2**: Persona detection accuracy ≥ 85% (validated against manually-labeled test dataset)
- **AC 1.2.2.3**: Hybrid persona support: Users can exhibit traits from multiple personas (e.g., "Student + Gamer")
- **AC 1.2.2.4**: Persona influences downstream recommendations:
  - App suggestions (15-25 apps per persona)
  - System settings (power plan, display settings, notifications)
  - Theme/wallpaper recommendations
- **AC 1.2.2.5**: User can explicitly correct persona if mis-classified ("Actually, I'm more of a gamer")
- **AC 1.2.2.6**: Persona stored in session state and used for telemetry analysis (anonymized)

Persona Taxonomy & Signals:

Persona	Detection Signals	Recommended Apps	System Settings
Gamer	Keywords: gaming, FPS, streaming, Discord, Twitch; mentions high-performance GPU; interest in customization	Steam, Epic Games, Discord, OBS Studio, GeForce Experience, AMD Adrenalin, Xbox Game Bar, MSI Afterburner	Game Mode ON, RGB lighting sync enabled, high-performance power plan, HDR auto-detect, G-Sync/FreeSync enabled
Student/Educator	Keywords: school, college, university, research, homework, assignments, study; mentions note-taking, collaboration	OneNote, Microsoft Teams, Zoom, Notion, Zotero, Adobe Acrobat Reader, Grammarly, Google Drive, Anki, Khan Academy	Blue light filter (night mode), battery saver enabled, cloud backup (OneDrive), Focus Assist during study hours, scheduled restart overnight
Professional/Business	Keywords: work, office, productivity, spreadsheets, presentations, video calls; mentions corporate apps, VPN	Microsoft Office Suite, Slack, Zoom, Adobe Acrobat, VPN clients (Cisco AnyConnect, FortiClient), LastPass, Grammarly, Dropbox, SharePoint	Balanced power plan, external monitor optimization, Windows Hello (fingerprint/face unlock), BitLocker encryption, scheduled backups
Content Creator	Keywords: video editing, photo editing, graphic design, music production, streaming, YouTube, social media	Adobe Creative Cloud (Premiere, Photoshop, Illustrator), DaVinci Resolve, OBS Studio, Audacity, GIMP, Blender, Canva, Handbrake	Color-accurate display calibration, high-performance power plan, external storage optimization, scratch disk configuration, GPU acceleration

Persona	Detection Signals	Recommended Apps	System Settings
			enabled
Casual/Home User	Keywords: email, web browsing, photos, videos, social media, shopping, family; no specific professional/gaming interests	Google Chrome/Firefox, Spotify, Netflix, VLC Media Player, Photos app, WhatsApp/Telegram, Microsoft Edge, PDF reader, antivirus	Balanced power plan, privacy-focused settings (tracking protection), parental controls (if kids mentioned), automatic updates enabled, dark mode

Technical Implementation Notes:

- **Intent Recognition**: Fine-tuned SLM with classification head for persona prediction
- **Feature Extraction**: Keyword spotting, entity recognition (apps, activities, devices mentioned)
- **Confidence Scoring**: Persona assigned only if confidence > 75%; otherwise, ask follow-up clarifying questions
- **Feedback Loop**: User corrections used to improve model via telemetry (opt-in)

Testing Scenarios:

1. Simulate gamer conversation → Verify persona = Gamer, recommended apps include Steam/Discord
2. Simulate student conversation → Verify persona = Student, settings include battery saver
3. Ambiguous signals → Verify Abhi asks clarifying question
4. User corrects persona → Verify persona updated, recommendations refreshed
5. Hybrid persona (e.g., Student + Creator) → Verify hybrid recommendations

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User Story 1.2.3 - Avoid Repetitive Questions

Priority: P0 (Must-Have)

As a user being interviewed by Abhi
I want the assistant to remember my previous answers and avoid asking the same question twice
So that the conversation feels natural and efficient

Acceptance Criteria:

- **AC 1.2.3.1**: Abhi maintains conversation context across all turns (context window: 4K tokens = ~50-60 exchanges)
- **AC 1.2.3.2**: If user mentions a fact (e.g., "I'm a video editor"), Abhi does not ask about occupation later
- **AC 1.2.3.3**: Follow-up questions build on previous answers (e.g., User: "I game a lot" → Abhi: "What genres do you play?")
- **AC 1.2.3.4**: User can proactively volunteer information (e.g., "I also need video conferencing apps") and Abhi integrates without re-asking
- **AC 1.2.3.5**: If information is unclear, Abhi politely asks for clarification once (not repeatedly)

Technical Implementation Notes:

- **Context Management**: Store conversation history in session state; prepend to each AI prompt as context
- **Entity Tracking**: Extract entities (occupation, interests, apps mentioned) and store in structured format
- **Duplicate Detection**: Before asking question, check if answer already in context
- **Dynamic Prompt Engineering**: Adjust prompt based on collected facts to avoid redundancy

Testing Scenarios:

1. User mentions occupation early → Verify no later question about "What do you do?"
2. User lists apps → Verify Abhi doesn't ask "What apps do you need?" again
3. User provides detailed answer → Verify follow-up is relevant, not repetitive
4. Edge case: User gives contradictory information → Verify Abhi politely seeks clarification

---

User Story 1.2.4 - Session Persistence

Priority: P0 (Must-Have)

As a user whose setup is interrupted (crash, power loss, intentional pause)
I want to resume from where I left off without starting over
So that I don't lose progress and can complete setup at my convenience

Acceptance Criteria:

- **AC 1.2.4.1**: Session state saved every 30 seconds to encrypted database
- **AC 1.2.4.2**: Session state includes:
  - Conversation history (all turns)
  - Detected persona and confidence score
  - User preferences and selections
  - Background task progress (app installs, updates)
  - Timestamp of last activity
- **AC 1.2.4.3**: On Abhi restart, if incomplete session detected:
  - Display recovery dialog: "Welcome back! Would you like to continue setup?" [Resume] [Start Over]
- **AC 1.2.4.4**: Resume option:
  - Loads previous session state
  - Displays last 5 conversation turns for context
  - AI greeting: "Let's pick up where we left off. You were [brief summary of last topic]."
- **AC 1.2.4.5**: Start Over option:
  - Clears session state
  - Begins fresh conversation
  - Logs telemetry for restart event
- **AC 1.2.4.6**: Session expires after 24 hours of inactivity (user must start over)

Technical Implementation Notes:

- **Persistence Layer**: Encrypted database at `%LOCALAPPDATA%\Abhi\session.db`

- **Serialization**: JSON serialization of session object with schema versioning
- **Auto-Save**: Background thread saves state every 30s; no UI blocking
- **Recovery UI**: Modal dialog on app start if `session.db` exists and `session_complete` = false

Testing Scenarios:

1. Mid-conversation crash → Restart Abhi → Verify recovery dialog, resume works
2. User manually closes Abhi → Restart → Verify recovery option available
3. Resume from saved state → Verify conversation context retained, no repeated questions
4. Start Over → Verify clean slate, previous state cleared
5. 24-hour expiration → Verify old session not recoverable

5.3 Epic 3 - Background Setup Tasks

Business Value: Maximize efficiency by parallelizing automation with user interaction
Priority: P0 (Must-Have)

User Story 1.3.1 - Silent Peripheral Updates

Priority: P1 (High Priority)

As a user with connected peripherals (monitor, keyboard, mouse, webcam, etc.)
I want Abhi to automatically update their firmware and drivers in the background
So that my devices work optimally without manual intervention

Acceptance Criteria:

- **AC 1.3.1.1**: On launch, Abhi enumerates all connected hardware via PnP subsystem (WMI `Win32_PnPEntity` query)
- **AC 1.3.1.2**: For each device, checks for available updates via:
    - Windows Update catalog (driver updates)
    - OEM update services (Dell Command)
    - Peripheral manufacturer APIs (Logitech, Razer, Corsair, etc. if partnerships established)

- **AC 1.3.1.3**: Updates download and install silently (no user prompts) during interview
- **AC 1.3.1.4**: Progress indicator in UI: "Setting up your devices..." with count (e.g., "3 of 5 devices updated")
- **AC 1.3.1.5**: If update requires device reconnection, display notification: "Please reconnect [Device Name] to complete setup"
- **AC 1.3.1.6**: Failed updates logged but do not block setup completion; user notified post-setup with retry option

Technical Implementation Notes:

- **Device Enumeration**: WMI query for `SELECT * FROM Win32_PnPEntity WHERE ConfigManagerErrorCode = 0` (working devices)
- **Update Check**: Parallel HTTP requests to vendor APIs with 10s timeout per device
- **Update Application**: Use vendor-provided silent installers (MSI, EXE with /silent flags)
- **Retry Logic**: 3 attempts per update with exponential backoff (1s, 2s, 4s)
- **Telemetry**: Log device type, update status (success/failure), duration

Testing Scenarios:

1. Connect monitor, keyboard, mouse → Verify updates detected and applied
2. No updates available → Verify progress indicates "up to date"
3. Update requires reconnection → Verify notification displayed
4. Failed update → Verify logged, setup continues, user notified post-completion
5. Performance: Updates do not degrade AI response latency

User Story 1.3.2 - Minimize Disruptions (Restart Management)

Priority: P0 (Must-Have)

As a user completing the setup interview
I want Abhi to avoid rebooting my PC during the conversation
So that I can complete setup without interruption

Acceptance Criteria:

- **AC 1.3.2.1**: OS and driver updates that require restart are queued, not executed immediately
- **AC 1.3.2.2**: User notified at end of interview: "To complete setup, we need to restart your PC. This will take about 2 minutes. [Restart Now] [Restart Later]"
- **AC 1.3.2.3**: "Restart Now" option:
  - Saves session state
  - Displays countdown timer (60 seconds with cancel option)
  - Initiates restart via `shutdown /r /t 60`
  - On next login, Abhi auto-launches to complete post-restart tasks
- **AC 1.3.2.4**: "Restart Later" option:
  - Schedules restart for user-selected time (next 1-24 hours)
  - Displays Windows notification 5 minutes before scheduled restart
  - User can postpone up to 3 times (max 72 hours total delay)
- **AC 1.3.2.5**: Critical security updates bypass delay (immediate notification: "Critical update requires restart")

Technical Implementation Notes:

- **Restart Detection**: Monitor `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\Auto Update\RebootRequired`
- **Scheduling**: Use Task Scheduler to create delayed restart task
- **Notification**: Windows Toast notification API (`Microsoft.Toolkit.Uwp.Notifications`)
- **Post-Restart Resume**: Task Scheduler trigger on logon to complete final setup steps

Testing Scenarios:

1. Update requires restart → Verify queued, not immediate
2. End of interview → Verify restart dialog appears if updates pending
3. Restart Now → Verify countdown, restart, post-restart resume
4. Restart Later → Verify scheduled task created, notification appears

5. Critical update → Verify immediate notification, cannot postpone

---

User Story 1.3.3 - Persona-Based App Installation

Priority: P0 (Must-Have)

As a user whose persona has been detected
I want Abhi to start installing recommended apps in the background (with my consent)
So that they're ready to use when setup completes

Acceptance Criteria:

- **AC 1.3.3.1**: After persona detection, Abhi presents recommended apps:
  - Display: Grid view with app icons, names, brief descriptions
  - Default: All apps selected (checkboxes pre-checked)
  - User can deselect any app
  - "Install Selected Apps" button
- **AC 1.3.3.2**: Clicking "Install Selected Apps" triggers background installation:
  - Progress bar for overall installation
  - Individual app status (Pending → Downloading → Installing → Installed / Failed)
  - Estimated time remaining
- **AC 1.3.3.3**: Installations run in parallel (max 3 simultaneous to avoid bandwidth saturation)
- **AC 1.3.3.4**: Installation methods:
  - Microsoft Store apps: `ms-windows-store:` protocol or Store COM API
  - Winget apps: `winget install --silent --accept-package-agreements`
  - Custom installers: MSI/EXE with silent flags
- **AC 1.3.3.5**: Failed installations do not block setup; logged for post-setup retry
- **AC 1.3.3.6**: User can continue conversation during installations (non-blocking)

Technical Implementation Notes:

- **App Catalog**: JSON configuration file mapping personas to app lists with metadata (name, description, source, install command)

- **Consent Tracking**: User's app selections stored in session state
- **Parallel Installation**: Thread pool with semaphore (max 3 concurrent)
- **Error Handling**: Retry failed installs once; log final status
- **Telemetry**: Log app selection, installation success/failure, duration per app

Testing Scenarios:

1. Persona detected → Verify recommended apps displayed
2. User selects subset of apps → Verify only selected apps installed
3. Installation progress → Verify accurate status updates, time estimates
4. Failed installation → Verify logged, setup continues, no crash
5. Performance: Installations do not block UI or degrade conversation latency

---

5.4 Epic 4 - Device Setup & Peripheral Guidance

Business Value: Comprehensive hardware configuration reduces user frustration and support calls
Priority: P1 (High Priority)

User Story 1.4.1 - Multi-Monitor Detection and Configuration

Priority: P1 (High Priority)

As a user with multiple monitors
I want Abhi to automatically detect and optimally configure my displays
So that I have a proper multi-monitor setup without manual adjustment

Acceptance Criteria:

- **AC 1.4.1.1**: On launch, Abhi detects all connected displays via Windows Display API (`GetDisplayConfigBufferSizes`, `QueryDisplayConfig`)
- **AC 1.4.1.2**: For each display, retrieves metadata:
    - Resolution, refresh rate, color depth
    - Physical dimensions (if available via EDID)
    - Connection type (HDMI, DisplayPort, USB-C)
    - Manufacturer and model

- **AC 1.4.1.3**: Combines detected peripheral information with user input:
    - Detects all connected peripherals (monitors, keyboard, mouse, webcam, speakers, etc.)
    - Prompts user to confirm or correct detected setup (e.g., "I see you have 2 monitors. Are they side-by-side or stacked?")
    - Gathers any additional peripheral details from user conversation
- **AC 1.4.1.4**: Creates visual render of physical setup:
    - Generates rough 2D/3D visualization showing desk layout with detected peripherals
    - Shows monitor positions, keyboard, mouse, and other devices in approximate spatial arrangement
    - Displays configuration summary (resolution, scaling, refresh rates, audio output device, etc.)
    - Asks user to confirm: "Does this look like your setup? [Yes - Apply Configuration] [No - Let me adjust]"
- **AC 1.4.1.5**: Proposes optimal configuration:
    - Primary display selection (usually highest resolution/refresh rate)
    - Display arrangement (left-to-right, top-to-bottom based on user confirmation)
    - Resolution and scaling recommendations (e.g., 100% for 1080p, 150% for 4K)
- **AC 1.4.1.6**: Displays visual preview of proposed arrangement with drag-to-adjust capability
- **AC 1.4.1.7**: User can:
    - Accept configuration (apply immediately)
    - Manually adjust arrangement (drag displays in preview)
    - Override resolution/scaling per display
- **AC 1.4.1.8**: After applying, displays "Test Configuration" dialog for 15 seconds:
    - "Can you see this message on all displays? [Yes] [No]"
    - If "No" or timeout, revert to previous configuration

Technical Implementation Notes:

- **Peripheral Detection**: Enumerate all PnP devices; categorize by device class (display, HID, audio, etc.)
- **Visual Render Generation**: Use 2D/3D rendering library to create desk setup visualization
    - Position peripherals based on detected configuration and user input
    - Use simple geometric shapes or icons to represent devices
    - Render in-app before making system configuration changes
- **Display Detection**: `EnumDisplayDevices`, `EnumDisplaySettings` Win32 APIs
- **Configuration Application**: `SetDisplayConfig` API with `SDC_APPLY | SDC_SAVE_TO_DATABASE` flags
- **Visual Preview**: Canvas or graphics API for display arrangement preview
- **Revert Logic**: Save previous configuration before applying; revert on failure or user request

Testing Scenarios:

1. Single monitor → Verify detected, visual render shows single display, optimal resolution applied
2. Dual monitors → Verify arrangement proposed, visual render accurate, primary display correct
3. Triple+ monitors → Verify complex arrangement supported, render shows all displays
4. User corrects peripheral positions → Verify visual render updates, configuration respects changes
5. User confirms render → Verify configuration applied only after confirmation
6. Test configuration timeout → Verify revert to previous settings

---

User Story 1.4.2 - Desk Setup Photo Analysis

Priority: P2 (Nice-to-Have)

As a user with multiple peripherals on my desk
I want to upload a photo of my desk setup so Abhi can detect devices and offer setup help
So that I don't have to manually describe each device

Acceptance Criteria:

- **AC 1.4.2.1**: Interview includes question: "Would you like to upload a photo of your desk? This helps me detect peripherals."
- **AC 1.4.2.2**: User uploads photo (drag-drop or file picker)
- **AC 1.4.2.3**: AI analyzes photo and detects:
  - Monitors (count, approximate size)
  - Peripherals (keyboard, mouse, webcam, speakers, microphone, headphones, external drives)
  - Docking station or USB hub
- **AC 1.4.2.4**: Detection accuracy ≥ 80% for common devices
- **AC 1.4.2.5**: Abhi displays detected devices: "I see you have [device list]. Would you like me to help set these up?"
- **AC 1.4.2.6**: For detected devices not yet connected, Abhi offers pairing guidance (especially for Bluetooth devices)

Technical Implementation Notes:

- **Image Analysis**: Vision API for on-device object detection
- **Object Classes**: Pre-trained model with 20+ peripheral classes
- **Fallback**: Cloud-based Computer Vision API (with user consent) for improved accuracy
- **Confidence Filtering**: Only report detections with confidence > 70%

Testing Scenarios:

1. Upload photo with monitor + keyboard + mouse → Verify all detected
2. Upload photo with webcam + speakers → Verify detected
3. Upload photo with Bluetooth headphones (not connected) → Verify pairing guidance offered
4. Upload photo with no peripherals → Verify graceful handling ("I don't see any additional devices")

5. Performance: Analysis completes in < 3 seconds

User Story 1.4.3 - Bluetooth Device Pairing Assistance

Priority: P1 (High Priority)

As a user with Bluetooth peripherals (headphones, keyboard, mouse)
I want Abhi to guide me through the pairing process
So that I can connect devices without navigating Windows settings manually

Acceptance Criteria:

- **AC 1.4.3.1**: Abhi detects Bluetooth adapter via WMI (`Win32_Bluetooth` query)
- **AC 1.4.3.2**: If Bluetooth adapter present, asks: "Do you have any Bluetooth devices to connect?"
- **AC 1.4.3.3**: If yes, displays guided pairing workflow:
    - Step 1: "Please put your device in pairing mode (usually by holding the power button for 3-5 seconds)"
    - Step 2: Abhi initiates Bluetooth discovery (`Windows.Devices.Bluetooth` API)
    - Step 3: Displays list of discovered devices with signal strength indicators
    - Step 4: User selects device → Abhi initiates pairing
    - Step 5: If PIN required, displays dialog for user to enter PIN
    - Step 6: Confirms successful connection
- **AC 1.4.3.4**: Supports multiple device pairing in sequence
- **AC 1.4.3.5**: If pairing fails, offers troubleshooting tips:
    - "Make sure device is in pairing mode"
    - "Try moving device closer to PC"
    - "Check device battery level"
    - [Retry] button

Technical Implementation Notes:

- **Bluetooth API**: `Windows.Devices.Bluetooth` and `Windows.Devices.Enumeration` namespaces

- **Device Discovery**: `DeviceInformation.FindAllAsync` with Bluetooth selector
- **Pairing**: `DevicePairingResult` object handles PIN/passkey scenarios
- **Error Handling**: Specific error messages for common pairing failures (timeout, wrong PIN, etc.)

Testing Scenarios:

1. Bluetooth headphones → Verify guided pairing, successful connection
2. Bluetooth keyboard → Verify PIN entry dialog if required
3. Multiple devices → Verify sequential pairing works
4. Pairing failure → Verify troubleshooting tips displayed, retry option
5. No Bluetooth adapter → Verify graceful skip of Bluetooth setup

---

5.5 Epic 5 - Data Migration Assistance

Business Value: Seamless data transfer is critical for user satisfaction and reduces abandonment
Priority: P0 (Must-Have)

User Story 1.5.1 - Detect Previous Setup

Priority: P0 (Must-Have)

As a user who previously owned a PC
I want Abhi to detect if I have data to migrate
So that I'm not left wondering how to transfer my files

Acceptance Criteria:

- **AC 1.5.1.1**: Interview includes question: "Are you upgrading from another PC or device?"
- **AC 1.5.1.2**: If yes, Abhi asks: "How would you like to transfer your files?" and presents options:
    - **Cloud Sync** (OneDrive, Google Drive, Dropbox)
    - **Network Transfer** (from PC on same Wi-Fi network)
    - **USB/External Drive** (copy files from connected drive)
    - **Skip** (I'll transfer manually later)

- **AC 1.5.1.3**: Each option includes brief explanation and estimated time
- **AC 1.5.1.4**: User's choice determines migration workflow

Technical Implementation Notes:

- **Decision Tree**: Store user's migration choice in session state to drive subsequent flows
- **Estimation**: Rough time estimates based on typical file sizes (Cloud: ~30 min, Network: ~15 min, USB: ~10 min)

Testing Scenarios:

1. User has old PC → Verify migration options displayed
2. User selects Cloud Sync → Verify guided to cloud setup workflow
3. User selects Network Transfer → Verify guided to network discovery
4. User selects USB → Verify guided to file browser
5. User selects Skip → Verify migration skipped, no further prompts

---

User Story 1.5.2 - Cloud Sync Setup

Priority: P0 (Must-Have)

As a user who wants to migrate files via cloud storage
I want Abhi to help me connect my cloud account and sync my files
So that my data is accessible on the new PC

Acceptance Criteria:

- **AC 1.5.2.1**: Abhi asks: "Which cloud service do you use?" [OneDrive] [Google Drive] [Dropbox] [Other]
- **AC 1.5.2.2**: For OneDrive:
    - Initiates OneDrive sign-in flow (OAuth via Windows Credential Manager)
    - Configures OneDrive folder locations (Desktop, Documents, Pictures)
    - Enables Files On-Demand
    - Starts initial sync (progress displayed)
- **AC 1.5.2.3**: For Google Drive:

    - Launches Google Drive installer if not present
    - Guides user through Google account authentication
    - Configures sync settings (selective sync for large accounts)
- **AC 1.5.2.4**: For Dropbox:
    - Launches Dropbox installer if not present
    - Guides user through Dropbox authentication
    - Configures sync settings
- **AC 1.5.2.5**: Displays sync progress:
    - Files synced count
    - Data transferred (MB/GB)
    - Estimated time remaining
    - Errors (if any) with troubleshooting tips
- **AC 1.5.2.6**: User can continue interview while sync runs in background

Technical Implementation Notes:

- **OneDrive Integration**: Windows Credential Manager for tokens, OneDrive Settings API for configuration
- **Third-Party Installers**: Silent install for Google Drive (`GoogleDriveSetup.exe /silent`) and Dropbox (`DropboxInstaller.exe /S`)
- **Sync Monitoring**: Query cloud client APIs for sync status, file counts
- **Error Handling**: Common issues (no internet, quota exceeded, authentication failure) → display specific error messages

Testing Scenarios:

1. OneDrive setup → Verify sign-in, folder config, sync starts
2. Google Drive setup → Verify installer download, authentication, sync
3. Dropbox setup → Verify installer, authentication, sync
4. Sync progress → Verify accurate file counts, data transferred
5. Sync error (no internet) → Verify error message, retry option

---

User Story 1.5.3 - Network Transfer (PC-to-PC)

Priority: P1 (High Priority)

As a user with an old PC on the same Wi-Fi network
I want Abhi to connect to my old PC and transfer files directly
So that I can migrate data without using cloud or USB drives

Acceptance Criteria:

- **AC 1.5.3.1**: Abhi displays: "Make sure your old PC is powered on and connected to the same Wi-Fi network."
- **AC 1.5.3.2**: Abhi scans network for Windows PCs via NetBIOS name broadcast and SMB discovery
- **AC 1.5.3.3**: Displays list of discovered PCs with:
  - Computer name
  - IP address
  - Last seen timestamp
- **AC 1.5.3.4**: User selects source PC
- **AC 1.5.3.5**: Prompts for source PC credentials (username/password) if required
- **AC 1.5.3.6**: Option to install small helper application on source PC to enumerate user data locations:
  - Helper app scans for standard user folders (Desktop, Documents, Pictures, Downloads, Videos, Music)
  - Helper app detects application-specific data folders (browser profiles, email data, custom app data)
  - Provides folder enumeration results back to Abhi on new PC
- **AC 1.5.3.7**: Displays folder browser for source PC:
  - Shows detected folders with sizes and file counts
  - Default selection: Desktop, Documents, Pictures, Downloads
  - User can select additional folders
  - Shows original path locations for reference

- **AC 1.5.3.8**: Initiates transfer with location mapping:
    - Migrates files to same relative locations as source PC where possible (e.g., C:\Users\Username\Documents → C:\Users\Username\Documents)
    - Creates necessary folder structures on target PC
    - Copies files using robocopy with progress reporting
    - Handles conflicts (duplicate files) → prompt user for action (Skip / Overwrite / Keep Both)
- **AC 1.5.3.9**: Transfer progress:
    - Files transferred count
    - Data transferred (MB/GB)
    - Transfer speed (MB/s)
    - Estimated time remaining

Technical Implementation Notes:

- **Helper Application**: Lightweight executable (<5MB) that runs on source PC to enumerate file system
    - Scans known user folder locations using Windows Known Folders API
    - Detects common application data paths
    - Returns JSON manifest of folders with metadata (path, size, file count)
    - Secure communication with Abhi via TLS-encrypted connection
- **Network Discovery**: WMI `Win32_ComputerSystem` query on remote PCs, NetBIOS enumeration
- **Location Mapping**: Preserve source PC folder structure; map C:\Users[OldUser]\ to C:\Users[NewUser]\
- **File Transfer**: Robocopy CLI (`robocopy \\source\path C:\Users\... /E /Z /R:3 /W:5 /MT:8`) for robust, resumable transfers
- **Progress Monitoring**: Parse robocopy stdout for file counts, bytes transferred
- **Credential Handling**: Windows Credential Manager to securely store source PC credentials

Testing Scenarios:

1. Old PC on same network → Verify discovered, credentials prompt
2. Select folders → Verify file browser, default selections
3. Transfer files → Verify robocopy execution, progress display
4. Duplicate file conflict → Verify prompt, user choice respected
5. Network interruption → Verify transfer resumes from checkpoint

---

User Story 1.5.4 - USB/External Drive Transfer

Priority: P1 (High Priority)

As a user who copied files to a USB drive from my old PC
I want Abhi to detect the drive and help me copy files to the new PC
So that I can complete migration without manual file explorer usage

Acceptance Criteria:

- **AC 1.5.4.1**: Abhi displays: "Please connect your USB drive or external hard drive containing your files."
- **AC 1.5.4.2**: Monitors for new removable drive connection (`WM_DEVICECHANGE` Windows message)
- **AC 1.5.4.3**: Once detected, displays drive info:
  - Drive letter
  - Drive label
  - Total size / free space
- **AC 1.5.4.4**: Displays folder browser for USB drive:
  - User selects folders/files to copy
  - Default: Copy entire drive contents
- **AC 1.5.4.5**: Prompts for destination: [Documents] [Desktop] [Pictures] [Custom Location]
- **AC 1.5.4.6**: Initiates copy:
  - Uses Windows Copy Engine for native experience
  - Progress dialog (Windows Task Dialog API)

    - Handles conflicts (same as network transfer)

- **AC 1.5.4.7**: Copy complete notification:
    - "Successfully copied [N] files ([X] GB)"
    - [Open Destination Folder] button

Technical Implementation Notes:

- **Drive Detection**: `WM_DEVICECHANGE` message handling, query `Win32_LogicalDisk` where `DriveType = 2` (removable)
- **File Copy**: `IFileOperation` COM interface for Windows-native copy with progress
- **Conflict Handling**: `IFileOperation::SetOperationFlags(FOF_RENAMEONCOLLISION)` for automatic rename on conflict

Testing Scenarios:

1. Connect USB drive → Verify detected, drive info displayed
2. Select folders → Verify file browser, selection works
3. Copy files → Verify Windows-native copy progress dialog
4. Duplicate file conflict → Verify handled per user preference or auto-rename
5. Copy complete → Verify success notification, open folder button works

---

5.6 Epic 6 - Session Management & Recovery

Business Value: Fault tolerance and persistence are critical for high completion rates
Priority: P0 (Must-Have)

(User Stories covered earlier in 1.2.4 - Session Persistence; additional scenarios included here)

User Story 1.6.1 - Crash Recovery

Priority: P0 (Must-Have)

As a user whose Abhi application crashes unexpectedly
I want the app to recover gracefully on restart
So that I don't lose my setup progress

Acceptance Criteria:

- **AC 1.6.1.1**: If Abhi crashes (unhandled exception, process termination), Windows Error Reporting captures crash dump
- **AC 1.6.1.2**: On next launch, Abhi detects incomplete session (as in 1.2.4)
- **AC 1.6.1.3**: Recovery dialog includes additional context: "Abhi encountered an error. Don't worry, your progress was saved."
- **AC 1.6.1.4**: User can:
  - Resume session (loads last saved state)
  - Report problem (opens feedback form with pre-filled crash data)
  - Start over
- **AC 1.6.1.5**: Background tasks (app installs, updates) resume from last checkpoint
- **AC 1.6.1.6**: Crash telemetry logged for analysis (exception type, stack trace, session ID)

Technical Implementation Notes:

- **Exception Handling**: Top-level `AppDomain.UnhandledException` handler saves session state before exit
- **Crash Reporting**: Windows Error Reporting integration (`WerRegisterFile` to attach session log)
- **Task Resume**: Check task status on restart; resume in-progress tasks

Testing Scenarios:

1. Simulate crash during conversation → Restart → Verify recovery dialog, resume works
2. Simulate crash during app install → Restart → Verify install resumes
3. Simulate crash after 24 hours → Verify session expired, must start over
4. Report problem → Verify feedback form opens with crash data

---

User Story 1.6.2 - User-Initiated Pause

Priority: P1 (High Priority)

As a user who needs to step away during setup
I want to pause and resume later
So that I can complete setup at my convenience

Acceptance Criteria:

- **AC 1.6.2.1**: UI includes [Pause Setup] button (always visible in header)
- **AC 1.6.2.2**: Clicking pause displays dialog: "You can resume setup anytime. Just click the Abhi icon in your Start Menu."
- **AC 1.6.2.3**: Pause saves session state and minimizes/closes Abhi
- **AC 1.6.2.4**: Start Menu shortcut remains available: "Resume PC Setup"
- **AC 1.6.2.5**: Background tasks continue even when paused (app installs, updates)
- **AC 1.6.2.6**: User can resume by:
    - Clicking Start Menu shortcut
    - Abhi auto-launches after 4 hours of inactivity (optional, configurable)

Technical Implementation Notes:

- **Pause Mechanism**: Save state, set `session_paused` flag, minimize window
- **Resume**: Check for paused session on app start, display resume prompt
- **Background Tasks**: Continue in background Windows Service (not dependent on UI process)

Testing Scenarios:

1. Click Pause → Verify dialog, app minimizes
2. Background tasks continue → Verify apps install, updates download
3. Resume from Start Menu → Verify seamless resume
4. Auto-resume after 4 hours → Verify notification, app launches

---

5.7 Epic 7 - Privacy, Consent, and Data Retention

Business Value: Trust and compliance are non-negotiable for consumer product
Priority: P0 (Must-Have)

User Story 1.7.1 - Clear Data Collection Disclosure

Priority: P0 (Must-Have)

As a privacy-conscious user
I want to understand what data Abhi collects and how it's used
So that I can make informed decisions about my privacy

Acceptance Criteria:

- **AC 1.7.1.1**: First-run screen includes privacy notice before interview begins:
  - Title: "Privacy First"
  - Content: "Abhi uses AI to personalize your setup. Here's what you need to know:"
    - Data collected: Device info, setup preferences, conversation summary
    - Processing: Primary processing on-device; optional cloud AI for complex questions
    - Storage: Encrypted locally; auto-deleted after 30 days
    - No sharing: Data not shared with third parties without consent
  - [Learn More] link to detailed privacy policy
  - [Accept] [Decline (Exit)]
- **AC 1.7.1.2**: Declining privacy notice exits Abhi (sets opt-out flag)
- **AC 1.7.1.3**: Privacy policy accessible anytime via Settings menu
- **AC 1.7.1.4**: User can review collected data: Settings → "View My Data" → displays JSON export of session data

Technical Implementation Notes:

- **Privacy Notice**: Full-screen modal, must be acknowledged before proceeding
- **Data Export**: JSON serialization of session database with human-readable formatting

Testing Scenarios:

1. First run → Verify privacy notice displayed, cannot proceed without acceptance
2. Decline → Verify Abhi exits, opt-out flag set

3. Learn More → Verify privacy policy opens in browser
4. View My Data → Verify JSON export accurate, complete

User Story 1.7.2 - Consent for High-Impact Actions

Priority: P0 (Must-Have)

As a user concerned about unintended changes to my system
I want Abhi to ask for my explicit approval before making important changes
So that I remain in control of my PC

Acceptance Criteria:

- **AC 1.7.2.1**: Before executing high-impact actions, Abhi displays consent dialog:
  - Action description (e.g., "Install Microsoft Office Suite")
  - Explanation of what will happen
  - Estimated time
  - [Approve] [Decline] buttons
- **AC 1.7.2.2**: High-impact actions include:
  - Installing applications
  - Modifying system settings (power plan, privacy settings, firewall)
  - Accessing cloud accounts (OneDrive, Google Drive)
  - Using cloud AI services (fallback LLM)
- **AC 1.7.2.3**: User can decline individual actions without aborting entire setup
- **AC 1.7.2.4**: Consent choices saved in session state; telemetry logged (anonymized)

Technical Implementation Notes:

- **Consent Dialog**: Modal dialog, non-dismissible without user choice
- **Action Registry**: JSON file mapping action types to consent requirement levels (low/medium/high)

Testing Scenarios:

1. App installation → Verify consent dialog, approve installs

2. System setting change → Verify consent dialog, decline skips
3. Cloud account access → Verify consent dialog, approve proceeds
4. Multiple actions → Verify each requires individual consent

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User Story 1.7.3 - Data Deletion

Priority: P0 (Must-Have)

As a user who completed setup
I want to delete all data Abhi collected about me
So that I have control over my privacy

Acceptance Criteria:

- **AC 1.7.3.1**: Settings menu includes "Delete My Setup Data" option
- **AC 1.7.3.2**: Clicking displays confirmation:
  - "This will permanently delete all data Abhi collected during setup. You cannot undo this action."
  - [Cancel] [Delete Data]
- **AC 1.7.3.3**: Confirming deletes:
  - Session database
  - Conversation history
  - Persona data
  - Temporary files
- **AC 1.7.3.4**: After deletion, Abhi treats device as new (can re-run setup if desired)
- **AC 1.7.3.5**: Automatic deletion after 30 days post-setup completion (configurable via Group Policy for enterprise)

Technical Implementation Notes:

- **Manual Deletion**: Delete database file, clear registry flags, remove temp directories
- **Auto-Deletion**: Task Scheduler task runs daily, deletes sessions older than 30 days
- **Verification**: Log deletion event, verify files removed

Testing Scenarios:

1. Manual deletion → Verify all data deleted, Abhi resets
2. Auto-deletion after 30 days → Verify task executes, old data purged
3. Re-run setup after deletion → Verify fresh experience, no previous data

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6. Go-to-Market Strategy

6.1 Target Markets & Channels

Primary Market: Consumer AI PCs sold through retail and online channels
Secondary Market: Enterprise IT deployments of AI PCs to knowledge workers

6.2 Positioning & Messaging

Product Positioning: "The first AI assistant that sets up your PC for you—personalized, automated, and effortless."

Key Messages by Audience:

Audience	Message	Supporting Points
Consumers	"Your new PC is ready to use in 15 minutes—no tech skills required"	• Conversational AI guides you • Automatic app installation • Hassle-free file transfer
IT Managers	"Reduce onboarding costs by 20% with AI-powered PC setup automation"	• Standardized deployment • Support ticket reduction • Telemetry insights
C-Suite	"Differentiate your AI PC portfolio with the industry's first intelligent onboarding experience"	• Category leadership • Customer satisfaction lift • Competitive differentiation

6.3 Marketing Collateral Requirements

For Launch:

1. **Product One-Pager** (PDF, 2 pages):
    - Overview, key features, benefits, specifications

    - Target: OEM decision-makers, retail partners
2. **Demo Video** (90 seconds):
    - Show end-to-end setup experience
    - Target: YouTube, social media, website
3. **Press Release**:
    - Announce OEM partnerships, availability
    - Target: Tech media (The Verge, Ars Technica, PCWorld)
4. **Comparison Sheet** (Excel/PDF):
    - Abhi vs. traditional OOBE vs. Apple Migration Assistant
    - Target: OEM product managers, competitive analysis
5. **Sales Deck** (PowerPoint, 20 slides):
    - Value proposition, demo, roadmap
    - Target: OEM sales teams

Post-Launch:

1. **Customer Success Stories** (Case studies, 4-6 pages each):
    - Real-world deployments with metrics
    - Target: Enterprise prospects, PR
2. **Webinars** (45 min, quarterly):
    - Deep dive on features, best practices
    - Target: IT administrators, partners
3. **Social Media Content**:
    - Tips, tricks, user testimonials
    - Target: Twitter/X, LinkedIn, YouTube

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7. Development Timeline & Milestones

7.1 Phased Delivery Plan

To be determined.

7.2 Detailed Milestone Schedule

To be determined.

7.3 Critical Path & Dependencies

Critical Path Activities (delays here impact launch):

1. AI model training and optimization (Q1-Q2)
2. OEM API integrations (Dell, HP, Lenovo) (Q3)
3. Security audit and compliance (Q4)
4. Manufacturing integration (Q1 2027)

Key Dependencies:

- Windows 11 25H2 API availability
- OEM partner API documentation and test environments
- Legal approval of privacy policy and EULA
- Third-party app partnerships (Microsoft, Adobe, etc.)

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8. Success Metrics & Analytics

8.1 Telemetry & Instrumentation

Data Collection Strategy: Opt-in telemetry with user consent, anonymized, GDPR-compliant

Key Events to Track:

Event Category	Specific Events	Purpose
Session Lifecycle	Session start, pause, resume, completion, abandonment	Measure completion rates, identify drop-off points
User Interactions	Button clicks, voice input usage, image uploads	Understand feature usage, UX optimization
AI	Response latency, intent accuracy,	Monitor AI quality, identify

Event Category	Specific Events	Purpose
Performance	persona detection confidence	improvement areas
Automation Tasks	App install success/failure, update status, migration progress	Track automation reliability
Errors	Crashes, exceptions, API failures	Prioritize bug fixes, improve stability

Analytics Dashboard: Real-time Power BI dashboard for product, engineering, and OEM partners

8.2 Success Criteria & Acceptance Thresholds

Pre-Launch (Beta Phase):

- Setup completion rate ≥ 80% (target: 90% at GA)
- Average setup time ≤ 15 minutes (target: 8-12 at GA)
- Crash rate ≤ 2% of sessions (target: < 0.5% at GA)
- CSAT score ≥ 4.0/5.0 (target: 4.5 at GA)

Post-Launch (GA, First 6 Months):

- 1M+ activations globally
- Setup completion rate ≥ 90%
- Support ticket reduction ≥ 20% (vs. traditional OOBE baseline)
- NPS (Net Promoter Score) ≥ 50

Long-Term (12+ Months):

- 10M+ cumulative activations
- Enterprise adoption in ≥ 100 organizations
- 3rd-party developer integrations (app partnerships) ≥ 50

8.3 A/B Testing & Experimentation

Continuous Improvement via Experimentation:

Test 1: Interview Length

- Hypothesis: Shorter interview (5-7 min) improves completion vs. longer (10-15 min)
- Metric: Completion rate, CSAT

- Rollout: 50/50 split, 10K users per cohort

Test 2: Voice vs. Text-Only

- Hypothesis: Multi-modal (voice + text) increases engagement vs. text-only
- Metric: Interaction depth, session duration, satisfaction
- Rollout: 50/50 split, 5K users per cohort

Test 3: Automated vs. User-Selected Apps

- Hypothesis: Fully automated app install (no user selection) increases satisfaction vs. user-selected
- Metric: Completion rate, post-setup app usage, CSAT
- Rollout: 70/30 split (70% automated, control), 20K users

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9. Risks & Mitigation

9.1 Technical Risks

Risk	Impact	Probability	Mitigation Strategy
AI Model Latency Exceeds Target (>2s)	High - Poor UX, low satisfaction	Medium	• Optimize model quantization • Implement aggressive caching • Fallback to simpler prompts
OEM API Integration Delays	High - Blocks beta launch	Medium	• Prioritize top OEM (Dell) • Build abstraction layer for future OEMs • Fallback to generic Windows APIs
Security Vulnerability Discovered	Critical - Blocks GA launch	Low	• Continuous security audits (quarterly) • Bug bounty program pre-launch • Rapid patching process
Platform Dependency (Windows 11)	Medium - Limits addressable market	Low	• Windows 10 backport (degraded features) • Clear minimum requirements in

Risk	Impact	Probability	Mitigation Strategy
			marketing

9.2 Business Risks

Risk	Impact	Probability	Mitigation Strategy
OEM Partners Delay Adoption	High - Reduces launch volume	Medium	• Incentivize early adoption (co-marketing, rev share) • Demonstrate ROI via pilot metrics • Direct-to-consumer option (Microsoft Store)
User Privacy Backlash	High - Reputational damage	Low	• Privacy-first design (on-device AI) • Transparent data practices • Easy opt-out and data deletion
Competitor (Apple, Google) Launches Similar	Medium - Market differentiation lost	Medium	• First-mover advantage • Deeper OEM partnerships • Continuous innovation roadmap

9.3 Operational Risks

Risk	Impact	Probability	Mitigation Strategy
Cloud Service Downtime (Telemetry, AI Fallback)	Medium - Degraded experience	Low	• Offline-first architecture • Graceful degradation • 99.9% SLA with cloud provider
Scaling Issues (1M+ Concurrent Users)	High - Poor performance at launch	Medium	• Load testing (simulate 2M users) • Auto-scaling infrastructure • CDN for static assets
Support Overwhelm at Launch	Medium - User frustration	Medium	• Comprehensive in-app help • Community forums • OEM support training program

10. Dependencies & Integrations

10.1 Internal Dependencies

Dependency	Team/Dept	Criticality	Status
AI Model Training Data	Data Science	High	Not Started
Legal Approval (Privacy Policy, EULA)	Legal	Critical	Not Started
Cloud Infrastructure Setup	DevOps	High	Not Started
Marketing Assets (Videos, Deck)	Marketing	Medium	Not Started
OEM Partnership Agreements	Business Development	Critical	Not Started

10.2 External Dependencies

Dependency	Provider	Criticality	Mitigation
OEM Update APIs (Dell)	OEM Partners	High	Engage early, document APIs, build adapters
Microsoft Store API Access	Microsoft	Medium	Already available, test integration
Cloud AI Service	Cloud Provider	Low	Optional feature, on-device primary
Third-Party App Installers	Various (Adobe, Google, etc.)	Medium	Verify silent install flags, maintain app catalog

11. Regulatory & Compliance

11.1 Privacy Regulations

GDPR (EU):

- Data minimization: Only collect setup-essential data
- Right to access: "View My Data" feature
- Right to deletion: Manual and auto-delete after 30 days
- Consent: Explicit opt-in for data collection and cloud services
- Data portability: JSON export of user data

CCPA (California):

- Notice at collection: First-run privacy notice

- Right to delete: "Delete My Data" feature
- Right to opt-out: Skip setup or disable telemetry
- No sale of personal information: Commitment in privacy policy

Other Regions: Compliance team to assess requirements in China, India, Brazil

11.2 Accessibility Standards

WCAG 2.1 Level AA:

- Keyboard navigation for all functions
- Screen reader support (Narrator, JAWS)
- Minimum contrast ratios (4.5:1 for text)
- Resizable text (up to 200%)
- Alternative text for all images

Validation: Third-party accessibility audit (Q4 2026)

11.3 Security Certifications

Planned Certifications:

- **ISO 27001** (Information Security Management): Q2 2027
- **SOC 2 Type II** (for cloud services): Q3 2027
- **Common Criteria EAL2** (for enterprise deployments): Q4 2027

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12. Appendices

12.1 Glossary

Term	Definition
Abhi	Product codename for AI-Based Helper Interface
AI PC	Personal computer with dedicated NPU for AI workloads (40+ TOPS)
NPU	Neural Processing Unit - specialized hardware for AI inference
OOBE	Out-of-Box Experience - Windows initial setup wizard
Persona	User classification (Gamer, Student, Professional, Creator, Casual)
SLM	Small Language Model - lightweight AI model for on-device inference
Session State	Saved progress of setup interview and automation tasks

Term	Definition
Telemetry	Anonymous usage data collected for product improvement

12.2 Reference Documents

1. **Windows 11 Developer Documentation**: docs.microsoft.com/windows
2. **GDPR Compliance Guide**: gdpr.eu
3. **WCAG 2.1 Guidelines**: w3.org/WAI/WCAG21/quickref
4. **Microsoft Store Submission Guide**: docs.microsoft.com/store

12.3 Revision History

Version	Date	Author	Changes
0.1	2025-12-15	Product Team	Initial draft
0.5	2026-01-02	Product Team	Added technical architecture, epics 1-3
1.0	2026-01-06	Product Team	Complete PRD with all sections, ready for review

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Approval Signatures

Role	Name	Signature	Date
Product Manager	[Name]	_____	__
Engineering Lead	[Name]	_____	__
Marketing Director	[Name]	_____	__
Legal Counsel	[Name]	_____	__
VP Product	[Name]	_____	__

Document End